

Question	Answer	Marks
5(a)	(electric) field equals (electric) potential gradient	M1
	reference to minus sign	A1
5(b)	- for potential to be zero, one potential must be positive and the other potential must be negative - for potential to be zero, the charges must have opposite sign - for field to be zero, the fields (due to X and Y) must be in opposite directions - for field to be zero, the charges must have the same sign - the signs of the charges cannot (simultaneously) be both the same and opposite (so not possible) <i>Any three points, 1 mark each</i>	B3
5(c)(i)	$V_X = (-) Q / 4\pi\epsilon_0 X$ and $V_Y = (-) 2Q / 4\pi\epsilon_0 Y$	C1
	$(V_X + V_Y = 0 \text{ so}) Q / 4\pi\epsilon_0 X = 2Q / 4\pi\epsilon_0 Y$ leading to $y = 2x$	A1
5(c)(ii)	$E_X = Q / 4\pi\epsilon_0 X^2$	A1
5(c)(iii)	$E_Y = 2Q / 4\pi\epsilon_0 (2x)^2$ $(= Q / 8\pi\epsilon_0 X^2)$	C1
	(opposite charges so fields in same direction so magnitudes add): $E = (Q / 4\pi\epsilon_0 X^2) + (Q / 8\pi\epsilon_0 X^2)$ $= 3Q / 8\pi\epsilon_0 X^2$	A1

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